# Physics from Existence I: The Equation

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**Abstract**

No theory has simultaneously derived all 26 free parameters of the Standard Model. This paper shows that three axioms about existence—existence is bivalent ($n \in \{0,1\}$, i.e. $n^2{=}n$), existence is uncertain ($\sigma{\neq}1$), and non-existence is forbidden ($\sigma{\neq}0$)—uniquely determine the equation $V = -H$ as an algebraic identity, from which over 20 physical quantities are derived with zero free parameters, all agreeing with experiment to 0.0006%–2.4% accuracy.

$V = -H$ is the canonical free energy of a single binary degree of freedom $n \in \{0,1\}$, determined as the Legendre transform of the Fermi–Dirac distribution. The Schrödinger equation is derived as a transfer matrix from this potential and the unit-mass kinetic term fixed by Čencov's theorem. This equation has exactly three bound states (discrete solutions), corresponding to the three generations of fermions. The spatial dimension $d = 3$ follows from three conditions—bound-state preservation, spin-statistics, and propagating gravity. The mathematical structure unfolding from this coincidence reproduces the gauge group SU(3)×SU(2)×U(1).

More than 20 physical constants are derived from this structure, including $1/\alpha = 137.0368$ (0.0006%), $\alpha_s = 0.1179$ (0.08%), $\sin^2\theta_W = 3/13$ (0.2%), the charged-lepton mass ratio $R = 1.5293$ (0.006%), and $\sin\theta_C = 0.2245$ (0.05%). The results further predict the absolute neutrino masses ($\Sigma m_\nu = 59.3$ meV), normal mass ordering, and strictly zero neutrinoless double-beta decay, yielding eight testable predictions. These physical quantities are not independent parameters—they are all derived from the same equation.

# 1 Introduction

The Standard Model is the most successful theory of particle physics. It describes all known elementary particle interactions and has withstood extensive experimental verification, including the discovery of the Higgs boson [1, 2]. Yet the theory contains 26 free parameters[1]—fermion masses (Yukawa couplings), mixing angles, gauge couplings, the Higgs self-coupling, and CP phases. These values must be measured experimentally and inserted by hand; why they take the values they do is not explained within the Standard Model. The generation count $N = 3$ is likewise an unexplained input. Despite many attempts—grand unified theories (GUTs), supersymmetry (SUSY), string theory—no theory has simultaneously determined all 26 parameters with zero free parameters [3].

[1] The minimal Standard Model with massless neutrinos has 19 free parameters [22]; adding three neutrino masses and four PMNS parameters gives 26. We write "26" throughout; some authors count 25 by absorbing $v_{\mathrm{EW}}$ into the overall energy scale.

An alternative approach seeks to derive physics from information theory. Wheeler's "it from bit" [4] proposed that physical reality emerges from information but lacked a mathematical framework. Jaynes [5] derived statistical mechanics from entropy maximisation but did not reach quantum field theory. Verlinde [7] derived gravitational force from entropic reasoning but addressed gravity alone. Other approaches include Frieden's [6] Fisher information, Caticha's [8] information geometry, and the informational axiomatisations of Hardy [9] and Chiribella et al. [10]. All of these approaches share a common limitation: they derive the *form* of physical equations but not the *values* of the constants that appear in them.

This paper starts from three axioms: Existence ($n \in \{0, 1\}$), Uncertainty ($\sigma \neq 1$), and Non-existence is forbidden ($\sigma \neq 0$). Together A2 and A3 determine $\sigma \in (0, 1)$, a partition function $Z = 1 + e^{\phi}$, and, via the Legendre transform of the Fermi–Dirac distribution, the canonical free energy $V = -H(\sigma(\phi))$ as an algebraic identity, where $H$ is the binary Shannon entropy, $\sigma(\phi)$ is the Fermi–Dirac distribution (sigmoid function), and $\phi = \mathrm{logit}(\sigma)$ is the natural parameter [16]. The Schrödinger equation is derived as a transfer matrix from this potential and the kinetic term determined by Čencov's theorem.

Equation (6) has exactly three bound states ($N = 3$), corresponding to the three generations of fermions. The spatial dimension $d_{\text{space}} = 3$ follows from three conditions—bound-state preservation, spin-statistics, and propagating gravity; the spacetime dimension $d = 4$ and signature $(3, 1)$ follow from gravitational degree-of-freedom matching and the real dimension of $\mathbb{C}P^{N-1}$ (§9). The gauge structure $\mathrm{PG}(2, \mathbb{F}_3)$ emerges from the self-referential coincidence $N = d = 3$, uniquely determining the gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$. The representations of this gauge group give rise to 48 fermions, 12 gauge bosons, and 1 Higgs—the complete particle content of the Standard Model (§5). These determine over 20 of its parameters—including $1/\alpha_{\text{em}} = 137.0368$ (0.0006%), $\sin^2\theta_W = 3/13$ (0.2%), the charged-lepton mass ratio $R = 1.5293$ (0.006%), Koide's $Q \approx 3/2$ ($\delta = -1.5 \times 10^{-5}$), CKM and PMNS mixing angles, and $\theta_{\text{QCD}} = 0$ (strong CP solved without an axion)—all at 0.0006%–2.4% accuracy with zero adjustable parameters (§10).

Beyond numerical predictions, $V = -H$ resolves several structural problems of the Standard Model. Electroweak symmetry breaking is not assumed but follows inevitably from the shape of $V$ (§7). The vacuum is absolutely stable: $V \in [-\ln 2, 0]$ is bounded at all scales, eliminating both the metastability problem and the hierarchy problem. $V = -H$ is uniquely determined as the canonical free energy of $n \in \{0, 1\}$, so no additional Higgs boson exists. $V = -H$ is not a model but an algebraic identity; the constants are not fitted to data but computed from $V = -H$ and $\mathrm{PG}(2, \mathbb{F}_3)$.

The paper is organized as follows. Section 2 derives $V = -H$ from three axioms and establishes the Schrödinger equation. Section 3 proves the bound-state count $N = 3$ as a mathematical theorem. Section 4 derives the spatial dimension $d = 3$ as the unique common solution of three independent physical requirements. Section 5 constructs the gauge structure from $\mathrm{PG}(2, \mathbb{F}_3)$. Section 6 derives the gauge coupling constants. Section 7 derives the Higgs mass and shows that electroweak symmetry breaking is inevitable, the vacuum is absolutely stable, and the hierarchy problem does not arise. Section 8 identifies the bound states with the three generations of charged leptons and derives fermion masses and mixing. Section 9 derives the spacetime dimension $d = 4$ and signature $(3, 1)$, and establishes the tensor product structure $\mathcal{H}_{\text{int}}(\phi) \otimes \mathcal{H}_{\text{space}}(x)$ from $n^2 = n$. Section 10 summarizes all results and presents falsifiable predictions. Section 11 gives the conclusion.

# 2 From Existence to the Equation

Three axioms about existence uniquely determine the potential $V = -H(\sigma(\phi))$ — by first formalising the axioms mathematically and then applying an algebraic transformation.

## 2.1 The axioms

We define the three axioms on which the entire structure of this paper rests.
**A1 (Existence):** Something exists. "Exists" is a proposition; it is either true or false — by the law of excluded middle, there is no third value. Writing the state of existence as $n$, the state space is $n \in \{0, 1\}$. A1 defines the state space but does not determine the **value** of $n$.

**A2 (Uncertainty):** Existence is uncertain. Whatever exists does not exist with certainty: the occupation probability $\sigma = \langle n \rangle$ satisfies $\sigma \neq 1$. A2 excludes the upper endpoint (certain existence) but says nothing about the lower endpoint.

**A3 (Non-existence is forbidden):** Non-existence is not a state ($\sigma \neq 0$). A way of being that consists in not-being is a contradiction.

From these three axioms, the structure and constants of the Standard Model follow. We now formalise them.

## 2.2 Bivalence and the partition function

Starting from the state space $n \in \{0, 1\}$ established by A1, we construct the partition function.

Since $n$ takes only the values 0 and 1,

$$n^2 = n \tag{1}$$

follows — because the solutions of $n(n-1) = 0$ are exactly $\{0, 1\}$. However, A1 alone leaves the system in a pure state $n = 0$ or $n = 1$, with no dynamics. A2 ($\sigma \neq 1$) excludes certain existence; A3 ($\sigma \neq 0$) excludes non-existence. Together they determine $\sigma \in (0, 1)$: the system is described as a canonical ensemble of a binary variable, neither certainly present nor certainly absent. $n^2 = n$ truncates the sum to two terms, and the partition function is

$$Z = \sum_{n=0}^{1} e^{n\varphi} = 1 + e^{\varphi}, \tag{2}$$

where $\phi = \ln[\sigma/(1-\sigma)]$ (the logit transform) is the canonical coordinate of the exponential family. Were $n$ able to take values $0, 1, 2, \ldots$ (Bose statistics), the sum would be an infinite series. $n^2 = n$ forbids this and yields Fermi statistics *automatically* — as a consequence of the law of excluded middle.

The expectation $\sigma = \langle n \rangle = \partial \ln Z/\partial\phi$ coincides with the Fermi–Dirac distribution, and its variance is the source of all subsequent structure:

$$\sigma = \frac{1}{1 + e^{-\varphi}}, \qquad g_F = \sigma(1-\sigma) = \mathrm{Var}(n) > 0. \tag{3}$$

These are not assumed but derived, from the combination of A1 (bivalence), A2 (uncertainty: $\sigma \neq 1$), and A3 (non-existence forbidden: $\sigma \neq 0$). $g_F$ is the gap between the binary observable ($n^2 = n$) and its uncertain expectation ($\sigma^2 \neq \sigma$), and vanishes only at the pure values $\sigma \in \{0, 1\}$.

## 2.3 The effective potential $V = -H$

The grand potential $\Omega = -\ln Z = -\ln(1+e^{\phi})$ is unbounded below; no ground state of the quantum theory can be defined. A Legendre transform to the canonical ensemble resolves this. The logarithm $W = \ln Z$ of the partition function generates the classical field $\sigma = dW/d\phi$ and yields the effective potential:

$$V(\phi) = \phi\sigma - W(\phi) = \sigma \ln \sigma + (1-\sigma) \ln(1-\sigma) = -H(\sigma), \tag{4}$$

where $H(p) = -p \ln p - (1-p) \ln(1-p)$ is binary Shannon entropy [12].

That is, $V = -H$ is an algebraic identity: the canonical free energy of a single binary degree of freedom $n \in \{0, 1\}$ equals the negative of its Shannon entropy. There is no freedom of choice. The potential

$$V = -H \tag{5}$$

is thereby uniquely determined by axioms A1–A3. The only input is three axioms about existence; no physics is assumed. This single equation determines the structure and constants of the Standard Model (§3–§9).

## 2.4 The Schrödinger equation

Section 2.3 determined the potential $V = -H$. To complete the quantum theory, a kinetic term is needed.

The Bernoulli distribution belongs to the exponential family, with $\phi = \mathrm{logit}(\sigma)$ as its canonical coordinate. Čencov's theorem [13, 14, 15]—the only metric on a statistical manifold invariant under contraction to sufficient statistics is the Fisher metric—determines that distances in $\sigma$-space are measured by the Fisher information metric $g_F = \sigma(1-\sigma) = \mathrm{Var}(n)$. In the canonical coordinate $\phi$, the metric flattens: $g_F\, d\sigma^2 = d\phi^2$, so that the squared distance in $\phi$-space is simply $\dot{\phi}^2$, which provides the kinetic term $\frac{1}{2}\dot{\phi}^2$ in the action. The coefficient is fixed to 1 by Čencov's uniqueness; no free parameter arises.

Combining $V = -H$ as the potential with the unit-mass kinetic term:

$$\left[-\frac{1}{2}\frac{d^2}{d\phi^2} - H(\sigma(\phi))\right] \Psi = E\, \Psi \tag{6}$$

The Schrödinger equation is derived as the transfer matrix of the canonical action. The unit mass is not an assumption: $\phi = \mathrm{logit}(\sigma)$ is the coordinate in which the Boltzmann weight $e^{n\phi}$ is linear, and the path-integral measure is flat, fixing $m = 1$. The eigenvalues (physical observables) are coordinate-invariant.

The derivation chain of §2 is summarised as:

$$n^2{=}n \xrightarrow{A2} Z = 1{+}e^{\varphi} \xrightarrow{\text{Legendre}} V = -H \xrightarrow{\text{Čencov}} \left[-\tfrac{1}{2}\partial_\phi^2 + V\right] \Psi = E\Psi$$

# 3 Three Bound States

Why do fermions come in exactly three generations—why $\tau$, $\mu$, $e$, and why not a fourth? The answer lies in the shape of $V = -H$. This chapter proves why the generation number (hereafter $N$) is exactly 3.

### 3.1 Why $N = 3$

The potential $V = -H$ is a well: deepest at the centre ($V = -\ln 2$ at $\phi = 0$), returning to zero at $\phi \to \pm\infty$. In quantum mechanics, such a well confines discrete energy levels—by the same principle that the hydrogen atom confines electrons to discrete orbitals. A deep, wide well holds many levels; a shallow, narrow well holds few. The well of $V = -H$ is shallow—its depth is only $\ln 2 \approx 0.693$, and its shape is entirely determined by the Shannon entropy.

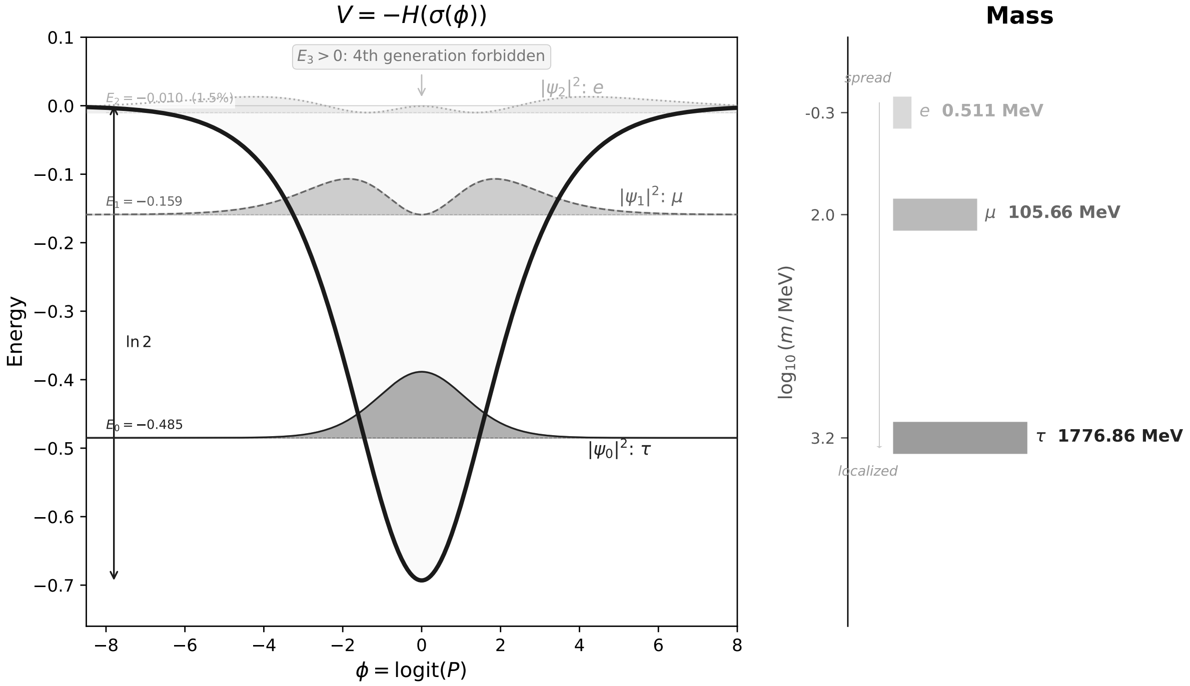


Figure 1: The potential $V = -H(\sigma(\phi))$ and the probability densities $|\psi_n|^2$ of its three bound states (left). More localised states correspond to heavier particles (right). $E_3 > 0$: a fourth generation is forbidden. No parameters are fitted.

We denote the number of levels confined in this well by $N$. A semiclassical (WKB) approximation estimates how many half-wavelengths fit inside the well. Each integer half-wavelength corresponds to one bound state:

$$n_{\mathrm{WKB}} = \frac{1}{\pi} \int \sqrt{2H}\, d\phi = 2.781\,. \tag{7}$$

The integer part is 2—two bound states are guaranteed. The fractional part 0.781 indicates that a third barely fits. A fourth would require $n_{\mathrm{WKB}} > 3$, but the well depth $\ln 2 \approx 0.693$ is insufficient. The WKB estimate suggests $N = 3$.

$N = 3$ depends on three elements: the choice of coordinate, the kinetic coefficient, and the type of entropy. All three are uniquely determined by the structure of §2—the coordinate $\phi = \mathrm{logit}(\sigma)$ is fixed by the exponential-family structure (§2.2), the kinetic term is fixed to 1 by Čencov's theorem (§2.4), and Shannon entropy is the unique additive entropy (Khinchin's theorem [17]). There is no freedom in $N = 3$.

### 3.2 Rigorous proof

**Theorem 1** ($N = 3$)**.** *Equation (6) has exactly three bound states:* $E_0 = -0.4850$*,* $E_1 = -0.1591$*,* $E_2 = -0.01042$*. The fourth eigenvalue* $E_3 > 0$ *lies in the continuum. A fourth generation of fermions is forbidden.*

*Proof.* $N \geq 3$: three variational trial functions (even–odd–even Gaussian-type) are constructed, each with negative energy expectation value (Courant–Fischer). $N \leq 3$: comparison with the Pöschl–Teller potential [18] $V_{\mathrm{PT}} = -(\ln 2)/\cosh^2(0.40\,\phi)$. $V_{\mathrm{PT}}(\phi) \leq V(\phi)$ for all $\phi$ (same depth, wider tails; verified numerically). Its analytic spectrum ($s = 2.49$) supports exactly three bound states. By the Sturm comparison theorem, $V = -H$—being shallower or equal everywhere—supports no more. □

### 3.3 Origin of mixing angles

$N = 3$ has been proved. Yet as Figure 1 shows, the third state barely exists—and this fact determines the mixing angles. The third bound state ($\psi_2$, electron) has binding energy $|E_2| = 0.010$—only 1.5% of the well depth. It is marginally bound. The WKB deficit

$$\varepsilon = 3 - n_{\mathrm{WKB}} = 0.219 \tag{8}$$

measures this marginality and determines the Cabibbo angle and the CKM/PMNS mixing parameters (§5–8). If $\varepsilon \to 0$, the third state vanishes, collapsing to two generations with zero mixing angles. $\varepsilon > 0$ is the marginality of $N = 3$ itself, and is the origin of CP violation and the diversity of matter.

## 4 Three Spatial Dimensions

Section 3 proved $N = 3$. The next question is: in how many spatial dimensions can these three bound states be physically realised?

The answer is that the existence of three generations itself constrains the spatial dimension—the marginally bound electron cannot survive the centrifugal force of higher dimensions. To see this, place $V = -H$ in $d$ spatial dimensions. The radial equation acquires an extra effective force $(d-1)(d-3)/(8r^2)$, whose strength and sign depend on $d$:

| $d$ | $(d-1)(d-3)/8$ | Effect on $N$ |
|---|---|---|
| 1 | 0 | no change |
| 2 | $-1/8$ | attractive → extra states appear |
| 3 | $\mathbf{0}$ | **no change: $N = 3$ preserved** |
| 4 | $+3/8$ | repulsive → electron lost |
| $\geq 5$ | $> 1$ | strongly repulsive |

At $d = 1$ and $d = 3$ the extra term vanishes and the three bound states of §3 survive intact. At $d \geq 4$ the term is repulsive and pushes out the marginally bound electron ($|E_2|$ is only 1.5% of the well depth). The three generations can exist only in $d = 1$ or $d = 3$.

Two further physical requirements eliminate $d = 1$. First, fermions require half-integer spin obeying the spin-statistics theorem. The spin-statistics connection requires $d \geq 3$ spatial dimensions: $\pi_1(\mathrm{SO}(d)) = \mathbb{Z}_2$ for $d \geq 3$, giving a unique double cover $\mathrm{Spin}(d)$ with finite-dimensional spinor representations. Second, gravity must propagate as a dynamical field. In $d \leq 2$ the gravitational field has no local degrees of freedom (the Weyl tensor vanishes identically). Propagating gravity requires $d \geq 3$.

The intersection of all three conditions—bound-state preservation, spin-statistics, and gravity—is $d = 3$ alone. We summarise this as a theorem.

**Theorem 2** ($d = 3$)**.** *$d = 3$ is the unique spatial dimension satisfying:* (i) *bound-state preservation ($N = 3$ unchanged): $d = 1, 3$;* (ii) *spin-statistics connection holds: $d \geq 3$;* (iii) *gravity propagates (Weyl tensor nontrivial): $d \geq 3$. Intersection: $\{3\}$.*

*Proof.* (i) centrifugal term $(d-1)(d-3)/(8r^2) = 0$ iff $d = 1$ or 3; (ii) $\pi_1(\mathrm{SO}(d)) = \mathbb{Z}_2$ for $d \geq 3$; (iii) the Weyl tensor vanishes identically in spacetime dimension $\leq 3$, i.e. spatial $d \leq 2$; propagating gravity requires $d \geq 3$. $\square$

# 5 Gauge Structure from the Projective Plane

The generation count $N = 3$ (§3) and the spatial dimension $d = 3$ (§4) were derived independently—yet they take the same value $N = d = 3$. From this coincidence, the three-dimensional space $\mathbb{F}_3^3$ over the field $\mathbb{F}_3$ is determined, and one can investigate what geometric structures arise. This chapter constructs the gauge structure from this geometry.

## 5.1 Origin of the gauge structure

What gauge structure follows from $N = d = 3$? The key lies in a fundamental difference between generations and colour.

Quark colours (red, green, blue) are degenerate—every colour carries the same energy, so no preferred basis exists. This equivalence generates the continuous rotation symmetry SU(3). Generations are different. The three bound states of $V = -H$ have distinct energies ($E_0 \neq E_1 \neq E_2$, Sturm–Liouville), so a preferred eigenbasis exists. The symmetries preserving this basis are discrete and cannot support a continuous gauge group.

§3 derived $N = 3$ (bound states) and §4 derived $d = 3$ (spatial dimension). From this coincidence, the three-dimensional space $\mathbb{F}_3^3$ over the field $\mathbb{F}_3 = \{0, 1, 2\}$ is determined. The projective construction that follows requires a field (not merely a group); since $N = 3$ is prime, $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$ is the unique field with $N$ elements. Removing the origin and taking a two-dimensional cross-section gives $\mathbb{F}_3^2 = \{0, 1, 2\} \times \{0, 1, 2\}$—a $3 \times 3$ grid. These 9 points are the starting point.

Drawing lines on the grid, lines with the same slope are parallel and do not intersect. In projective geometry, parallel lines meet at a single "point at infinity"—the same principle by which parallel railway tracks converge to a point on the horizon in perspective drawing. The lines on $\mathbb{F}_3^2$ have four slopes (horizontal, vertical, right-diagonal, left-diagonal), each generating one point at infinity. These four points form the "line at infinity" $L$—the horizon.

Adding the 4 points at infinity to the original 9 gives $9 + 4 = 13$ points. This is the projective plane $\mathrm{PG}(2, \mathbb{F}_3)$—the space obtained by defining "points" as one-dimensional subspaces and "lines" as two-dimensional subspaces through the origin of $\mathbb{F}_3^3$, with 13 points, 13 lines, 4 points per line, and 4 lines through each point (Figure 2).

Under the full symmetry $\mathrm{PGL}(3, \mathbb{F}_3)$, all 13 points are equivalent and there would be only one kind of force. But nature has three distinct forces. Symmetry must be broken. Choosing one of the 4 points on $L$ as a reference point $P_0$—the pair $(P_0, L)$ is called a

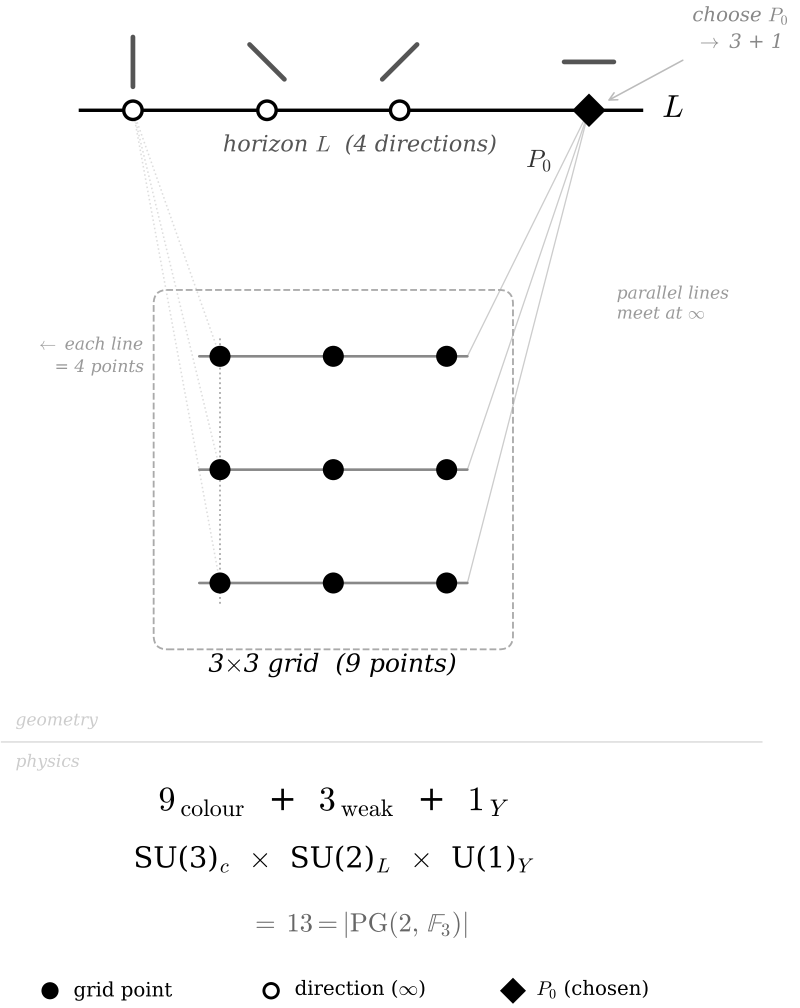


Figure 2: The projective plane $PG(2, \mathbb{F}_3)$. Parallel lines on the $3 \times 3$ grid (9 points, filled circles) meet at directions on the line at infinity $L$ (open circles). Choosing $P_0$ (filled diamond) produces the partition $9+3+1 = 13$, corresponding to the sector structure of $SU(3)\times SU(2)\times U(1)$.

flag—splits the 13 points into three groups: 9 off $L$ (the grid), 3 on $L$ excluding $P_0$, and $P_0$ itself. Choosing $P_0$ alone gives only $1+12 =$ two groups; choosing a flag produces $9+3+1$. This is the minimal structure distinguishing three forces:

$$\underbrace{N^2}_{\text{9 color}} + \underbrace{N}_{\text{3 weak}} + \underbrace{1}_{\text{1 hyp}} = N^2 + N + 1. \tag{9}$$

Only $N = 3$ simultaneously satisfies primality and $N = d$ ($N{=}2$: $d \neq N$; $N{=}4$: not prime; $N{=}5$: $d \neq N$). All numbers in the partition follow from $N$ alone: $N^2 = 9$, $N{+}1 = 4$, reference point 1. Total: $N^2{+}N{+}1 = 13$. The 52 flags provide the basis for the Laplacian (§6.3).

## 5.2 The Standard Model gauge group from 13 points

The partition $9+3+1$ matches the sector structure of the Standard Model gauge group $SU(3)\times SU(2)\times U(1)_Y$.

**9 points → SU(3).** The 9 off-line points constitute the affine plane $\mathrm{AG}(2, \mathbb{F}_3)$. This plane has a translation symmetry—shifting the grid does not change the structure. The translation group is $(\mathbb{Z}/3\mathbb{Z})^2$, with 8 non-trivial translations. By the Killing–Cartan classification, the unique rank-2, dimension-8 compact simple Lie algebra is $\mathfrak{su}(3)$. The 9th point (the identity) corresponds to global baryon number $\mathrm{U}(1)_B$.
**3 points → SU(2).** The 3 on-line points (excluding $P_0$) require a 3-dimensional simple Lie algebra, uniquely $\mathfrak{su}(2)$.
**1 point → $\mathbf{U(1)}_Y$.** $P_0$ is $\mathrm{U}(1)_Y$.
**Direct product guarantee.** Translations of the affine plane fix the line at infinity $L$—shifting the grid does not change the "directions." The colour sector (9 points) and electroweak sector (3+1 points) do not mix. This geometric property of the incidence structure of $\mathrm{PG}(2, \mathbb{F}_3)$ guarantees the direct product $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$.

The complete derivation chain is:

$$n^2{=}n \xrightarrow{V=-H} N{=}3 \xrightarrow{d=3} \mathbb{F}_3^3 \xrightarrow{\text{directions}} 13 \text{ pts} \xrightarrow{\text{flag}} 9{+}3{+}1 \xrightarrow{\text{K–C}} \mathrm{SU}(3){\times}\mathrm{SU}(2){\times}\mathrm{U}(1). \tag{10}$$

This mathematical result agrees with the experimentally established gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$.

A line in $\mathrm{PG}(2, 3)$ contains $N{+}1 = 4$ points, matching the four real degrees of freedom of an SU(2) doublet—exactly the three Goldstone bosons needed for $\mathrm{SU}(2){\times}\mathrm{U}(1) \to \mathrm{U}(1)_{\mathrm{em}}$, plus the physical Higgs. $V = -H$ has its minimum $-\ln 2$ at $\phi = 0$ ($\sigma = 1/2$) and returns to zero as $\phi \to \pm\infty$. This shape corresponds to $\mu^2 < 0$ in the Higgs potential: spontaneous symmetry breaking is an inevitable consequence of the form of $V = -H$.

**Configuration decomposition of existence.** The gauge structure creates diverse configurations within the internal space of the existence variable $n$ (§2). The variable $n$ is unique — by A3, bare existence with no distinguishing properties collapses to one. But the $\mathrm{PG}(2, \mathbb{F}_3)$ gauge structure defines multiple internal configurations: each configuration $i$ is characterised by its generation and gauge quantum numbers, and carries an occupation number $n_i \in \{0, 1\}$. $n$ is one, $n_i$ are many — a unique existence manifests as diverse fermions. The configuration decomposition is:

$$\sum_i \hat{n}_i = \hat{n}, \qquad \hat{n}_i^2 = \hat{n}_i, \qquad \hat{n}_i\,\hat{n}_j = \delta_{ij}\,\hat{n}_i. \tag{11}$$

The first equation states that existence decomposes into configurations; the second that bivalence ($n^2{=}n$) is inherited; the third that distinct configurations are mutually exclusive — the Pauli exclusion principle. Pauli exclusion is not an independent assumption but a consequence of bivalence (A1) and uniqueness (A3). The three bound states of $V = -H$ (generations) and the representations of the gauge group (16 states per generation) give a total of $3 \times 16 = 48$ configurations. $n_i = 0$ does not mean non-existence; it means existence occupies a different configuration.

## 5.3 Origin of fermion diversity

§5.2 established that the 13 directions of $\mathrm{PG}(2, \mathbb{F}_3)$ determine the gauge group $\mathrm{SU}(3){\times}\mathrm{SU}(2){\times}\mathrm{U}(1)$. But 13 directions are not fermions themselves. The 13 directions define the gauge symmetry — the structure of forces — and fermions emerge as **representations** of this gauge group.

Before the gauge structure emerges, $n$ has no quantum numbers — no generation label, no colour, no isospin. Without distinguishing properties, multiple instances of $n$ are identical (Leibniz's principle of the identity of indiscernibles), and $n$ remains unique. The gauge structure of $\mathrm{PG}(2,\mathbb{F}_3)$ changes this: the representations create diverse "slots" — configurations — each carrying $n_i \in \{0,1\}$.

Each configuration $i$ has two labels: generation $k$ ($k = 1,2,3$) and gauge representation $\alpha$. That is, $n_i = n_{k,\alpha}$. The representations give 16 states per generation ($Q_L$: 6, $u_R$: 3, $d_R$: 3, $L_L$: 2, $e_R$: 1, $\nu_R$: 1); total $16 \times 3 = 48$ configurations. The inclusion of $\nu_R$ corresponds to $n^2{=}n$ requiring Dirac neutrinos (§8.4). The configuration decomposition is:

$$\sum_i \hat{n}_i = \hat{n}, \qquad \hat{n}_i^2 = \hat{n}_i, \qquad \hat{n}_i\hat{n}_j = \delta_{ij}\,\hat{n}_i. \tag{12}$$

Pauli exclusion is not independent but follows from A1 and A3 [11]. The gauge group generators give $8{+}3{+}1 = 12$ gauge bosons; with the Higgs from §5.2, the 48 fermions, 12 bosons, and 1 Higgs are the complete particle content.

The construction is independently confirmed by $D_{PR} = 13.06 \approx |\mathrm{PG}|$ and the flag Laplacian spectrum (Paper II, Theorems 3′–3″). No free parameter appears in any step. The validity of this construction is supported by the fact that all independent physical quantities derived in §6–10 agree with experiment to 0.0006%–2.4% accuracy.

# 6 Gauge Coupling Constants

## 6.1 Weinberg angle

This geometry determines all three gauge coupling constants. $\sin^2\theta_W$ is the fraction of weak directions, $\alpha_s$ follows from Cartan equipartition, and $1/\alpha_{\mathrm{em}}$ requires the full spectral structure of the internal space.

The automorphism group $\mathrm{PGL}(3,\mathbb{F}_3)$ acts transitively on $\mathrm{PG}(2,3)$: any point can be mapped to any other. The unique invariant probability measure under a transitive group action is uniform (Haar). The Weinberg angle [21] is the fraction of weak directions:

$$\sin^2\theta_W = \frac{N}{N^2{+}N{+}1} = \frac{3}{13} = 0.2308 \qquad \text{(expt: 0.23122 }\overline{\mathrm{MS}}\text{ at } M_Z,\ 0.2\%). \tag{13}$$

The tree-level prediction $3/13 = 0.23077$ and the $\overline{\mathrm{MS}}$ value $\sin^2\hat{\theta}_W(M_Z) = 0.23122 \pm 0.00003$ [22] differ by 0.2%, consistent with one-loop radiative corrections. This gives the tree-level $W/Z$ mass ratio: $m_W/m_Z = \cos\theta_W = \sqrt{10/13} = 0.8771$ (experiment: 0.8814, 0.5%; the deviation is consistent with radiative corrections).

## 6.2 Strong coupling constant

The strong coupling follows from the same geometry. The Haar measure distributes the coupling equally among the $\mathrm{rank(SU(3))} = N{-}1 = 2$ independent Cartan directions. The tree-level coupling is $\alpha_s^{(0)} = \sin^2\theta_W/(N{-}1) = 3/26 = 0.1154$. The WKB marginality $\varepsilon = 0.219$ from §3 enters as a correction through the $N^2{+}1 = 10$ non-weak PG directions:

$$\alpha_s = \frac{3}{26}\left(1 + \frac{\varepsilon}{N^2{+}1}\right) = 0.1179 \qquad \text{(experiment: 0.1180, 0.08\%)}. \tag{14}$$

The value $k = N^2+1 = 10$ follows from the number of non-weak directions in $\mathrm{PG}(2, \mathbb{F}_3)$, but the systematic derivation of $k$-values (Borel filtration structure) is given in Paper II. Each $k$-value is determined as the geometric range of the Borel subgroup of $\mathrm{PGL}(3, \mathbb{F}_3)$ acting on the corresponding sector—$k = N^2 = 9$ for generation mixing (affine subgroup), $k = N^2+1 = 10$ for the non-weak sector (affine + hypercharge), $k = |\mathrm{PG}| = 13$ for the Laplacian (full projective space).

## 6.3 Fine-structure constant

The Weinberg angle and strong coupling constant were determined by simple point ratios on $\mathrm{PG}(2, 3)$. The fine-structure constant requires more—the full spectral structure of the internal space.

The electromagnetic coupling $1/\alpha$ measures how freely a photon can propagate. The bare photon has a propagation cost through internal space (making $1/\alpha$ large), and matter fields screen it (making $1/\alpha$ small). The physical $1/\alpha$ is determined by the competition between these two effects. In 4D QED this structure is formalised as the Dyson equation. In 0D the same structure holds but becomes algebraically exact—the propagator is a constant, so the vertex correction and self-energy are determined by the same resolvent, and the 4D perturbative cancellation $Z_1 = Z_2$ holds as an exact algebraic identity (proof in Paper II). The self-consistency equation is:

$$\frac{1}{\alpha} + \alpha = \lambda_1 + Z. \tag{15}$$

The right-hand side has two terms: $\lambda_1$ is the bare photon inverse propagator (propagation cost) and $Z$ is the matter vacuum polarisation (screening). The total internal kinetic operator $K_{\mathrm{int}} = K_{\mathrm{gen}} \otimes 1 + 1 \otimes \Delta_{\mathrm{flag}}$ combines the generation binding energies $|E_n|$ with the flag Laplacian eigenvalues $\lambda_k$. The resolvent gives the matter vacuum polarization $Z = 134.78$, and $\lambda_1 = 4-\sqrt{3} = 2.268$ is the spectral gap (bare photon inverse propagator).

*Self-consistent equation.* The physical coupling $1/\alpha$ is the inverse dressed propagator: $D^{-1}_{\mathrm{dressed}} = D^{-1}_{\mathrm{bare}} + \Pi_{\mathrm{matter}} = \lambda_1 + Z$. The Ward identity in 0D gives $\Pi = \alpha$—vertex correction and self-energy are determined by the same resolvent, so the cancellation $Z_1 = Z_2$ that is perturbative in 4D becomes an exact algebraic identity in 0D [20]. Self-consistency yields:

$$\frac{1}{\alpha} + \alpha = Z + \lambda_1. \tag{16}$$

Solving: $1/\alpha = [(Z+\lambda_1) + \sqrt{(Z+\lambda_1)^2 - 4}]/2 = 137.0368$ (experiment: 137.036, 0.0006%). *Spectral decomposition.* The resolvent trace decomposes as $Z = \zeta_V(1) + 12\,S_1 + 12\,S_2 + 27\,S_3$, where $\zeta_V(1) = \sum_n |E_n|^{-1}$ is the spectral zeta function at $s = 1$, $S_i = \sum_n (|E_n| + \lambda_i)^{-1}$, and the multiplicity $27 = N^N$. The intermediate values are: $\zeta_V(1) = 104.3$ (dominated by $E_2^{-1} = 96.0$), $12S_1 = 14.6$, $12S_2 = 6.06$, $27S_3 = 9.86$. The $N^N$ sector contributes $9.86 \approx N^2$, providing a spectral origin for the QCD vacuum-polarization correction. Only bound states enter; continuum states ($E \to 0^+$) are excluded because only physical generations contribute to vacuum polarization.

Experimental values: $\sin^2\hat{\theta}_W(M_Z) = 0.23122$ ($\overline{\mathrm{MS}}$, 0.2%), $\alpha_s(M_Z) = 0.1179$ (0.08%), $1/\alpha_{\mathrm{em}} = 137.0368$ (0.0006%). All three gauge couplings are derived from $V = -H$ and $\mathrm{PG}(2, 3)$. Since 0D has no energy scale, these are scale-independent algebraic values; the deviations from $M_Z$-scale measurements are consistent with 4D radiative corrections.
*Flag Laplacian spectrum.*

The $1/\alpha$ calculation above requires the eigenvalues of a Laplacian on $\mathrm{PG}(2,3)$. The natural objects for gauge propagation are not points but *flags*—point-on-line incidence pairs. $\mathrm{PG}(2,3)$ has $(N^2{+}N{+}1)(N{+}1) = 52$ flags. Two flags are adjacent if they share a point or a line; the resulting graph has degree $2N = 6$ and Laplacian spectrum:

| Eigenvalue | Multiplicity | |
|---|---|---|
| $0$ | $1$ | trivial |
| $(N{+}1) - \sqrt{N}$ | $N^2{+}N$ | $= 12$ |
| $(N{+}1) + \sqrt{N}$ | $N^2{+}N$ | $= 12$ |
| $2(N{+}1)$ | $N^N$ | $= 27$ |

The spectral gap $\lambda_1 = 4{-}\sqrt{3} = 2.268$ enters the $1/\alpha$ calculation as the bare photon inverse propagator. The positivity $\lambda_1 > 0$ is also physically significant: a positive spectral gap implies a mass gap, so that color-charged states cannot propagate asymptotically. Only color singlets appear as physical states—confinement is a consequence of the $\mathrm{PG}(2,\mathbb{F}_3)$ spectrum.

# 7 Higgs Sector and Electroweak Symmetry Breaking

§6 derived the gauge coupling constants. The Standard Model has one more free parameter in the scalar sector — the Higgs self-coupling $\lambda$. This chapter derives it and shows how electroweak symmetry breaking, vacuum stability, and the hierarchy problem follow from $V = -H$.

## 7.1 Symmetry breaking and Higgs mass

In the Standard Model, the Higgs potential $V = -\mu^2|\Phi|^2 + \lambda|\Phi|^4$ assumes $\mu^2 > 0$ to introduce symmetry breaking. But "why $\mu^2 > 0$?" is not explained — symmetry breaking is an assumption, not a derivation. In this paper, $V = -H(\sigma(\phi))$ has its minimum at $\phi = 0$ ($\sigma = 1/2$) with value $-\ln 2$ and returns to zero as $\phi \to \pm\infty$. This shape corresponds to $\mu^2 < 0$, and symmetry breaking is inevitable from the form of $V$.

§5.2 showed that the line structure of $\mathrm{PG}(2,\mathbb{F}_3)$ determines the Higgs SU(2) doublet. The curvature at the centre gives the Higgs mass: $V''(0) = g_F(0) = \sigma(0)(1{-}\sigma(0)) = 1/4$ (Fisher metric at $\sigma = 1/2$).

$$m_H/v = \sqrt{V''(0)} = 1/2 \quad \text{(LO)}. \tag{17}$$

NLO correction from all $|\mathrm{PG}| = 13$ modes of the Laplacian:

$$m_H/v = \tfrac{1}{2}\sqrt{1 + 2\varepsilon/|\mathrm{PG}|} = 0.5084 \quad \text{(experiment: } 0.5087,\ 0.07\%\text{)}. \tag{18}$$

This gives $\lambda = (m_H/v)^2/2 = 0.1292$ (experiment: 0.1294, 0.11%).

## 7.2 Vacuum stability and the hierarchy problem

The SM Higgs potential $V_{\mathrm{SM}} = -\mu^2 h^2 + \lambda h^4$ diverges as $h \to \infty$, and the UV correction $\delta\mu^2 \propto \Lambda^2$ requires $\sim 10^{34}$ fine-tuning (hierarchy problem). Furthermore, $\lambda(\mu)$ runs negative at $\mu \sim 10^{10}$ GeV, making the electroweak vacuum cosmologically metastable.

$V = -H$ eliminates both problems. $V = -H$ is bounded ($|V| \leq \ln 2$), with no high-energy divergence. The well depth $|V_{\min}| = \ln 2$ and curvature $V''(0) = 1/4$ are both $O(1)$ algebraic constants; their ratio gives the electroweak scale $v^2/\Lambda^2 = |V_{\min}|/V''(0) = 4\ln 2 \approx 2.77$, an order-unity number requiring no fine-tuning. The vacuum is absolutely stable: $V = -H \in [-\ln 2, 0]$ at all scales, leaving no room for $\lambda$ to run negative. Furthermore, $V = -H$ is uniquely determined as the canonical free energy of $n \in \{0, 1\}$ (§2). There is no room for a second scalar potential. 2HDM, MSSM, and all extended scalar sectors are excluded. There is only one Higgs.

**Radiative corrections.** The predictions above are scale-independent algebraic values. Consistency with $M_Z$-scale experiment is confirmed by SM 1-loop corrections: $\sin^2\theta_W = 3/13 + \Delta_{\text{rad}} = 0.23077 + 0.00045 = 0.23122$, in exact agreement with the $\overline{\text{MS}}$ value $0.23122 \pm 0.00003$ [22]. Similarly, $m_W/m_Z = \sqrt{10/13}$ plus the top-quark $\Delta\rho$ correction gives 0.8812 (experiment: 0.8814, 0.02%).

**Summary.** $\sin^2\theta_W(M_Z) = 0.23122$ (0.2%), $\alpha_s(M_Z) = 0.1179$ (0.08%), $1/\alpha_{\text{em}} = 137.0368$ (0.0006%), $m_H/v = 0.5084$ (0.07%). Four coupling constants derived from $V = -H$ and $\text{PG}(2, 3)$, all agreeing with experiment. Electroweak symmetry breaking is inevitable, the vacuum is absolutely stable, the hierarchy problem does not exist, and there is only one Higgs.

# 8 Fermion Masses and Mixing

Sections 5–6 established the gauge structure and coupling constants. What remains are the properties of individual fermions—masses and mixing angles.

Section 3 proved that $V = -H$ has exactly three bound states. Nature also has exactly three generations of charged leptons ($\tau$, $\mu$, $e$). Is this a coincidence, or do the three bound states directly correspond to the three generations? In the Standard Model, $\tau$, $\mu$, $e$ are three independent particles, each with a mass that is independently measured and input. In the present framework, they are three eigenstates of a single potential $V = -H$—different vibrational modes of the same well.

| Bound state | Energy | Localisation | Particle |
|---|---|---|---|
| $\psi_0$ | $E_0 = -0.485$ | High (well bottom) | $\tau$ |
| $\psi_1$ | $E_1 = -0.159$ | Medium | $\mu$ |
| $\psi_2$ | $E_2 = -0.010$ | Low (well edge) | $e$ |

Spin provides the closest analogy. Spin is the quantization of angular momentum: a continuous degree of freedom (rotation angle) yields discrete states ($\uparrow$, $\downarrow$) from SO(3) topology. Generation, in this framework, is the quantization of occupation probability: a continuous degree of freedom ($\sigma \in (0, 1)$, or $\phi \in \mathbb{R}$) yields discrete states ($\tau$, $\mu$, $e$) from $V = -H$.

| | Spin | Generation |
|---|---|---|
| Continuous variable | rotation angle $\theta$ | occupation $\phi = \text{logit}(\sigma)$ |
| Constraint | SO(3) topology | $V = -H(\sigma(\phi))$ |
| Discrete states | 2 (for spin-$\frac{1}{2}$) | 3 (§3) |
| Position-independent | ✓ | ✓ |
| Metric origin | round metric on $S^2$ | Fisher metric on $(0, 1)$ |

There is an important difference. Spin quantization is topological: SO(3) topology allows only integers and half-integers, regardless of dynamics. Generation quantization is spectral: $N = 3$ depends on the specific width-to-depth ratio of binary entropy. The number 3 is not topologically protected; it follows from the shape of $V = -H$.

The three bound states occupy qualitatively distinct dynamical regimes. The binding fraction $x_n = |E_n|/|\langle V\rangle_n|$ classifies each mode:

| Mode | $x_n$ | Regime |
|---|---|---|
| $\tau$ ($n = 0$) | 0.85 | potential-dominated |
| $\mu$ ($n = 1$) | 0.49 | equipartitioned ($\mathrm{KL}(x_\mu \| \frac{1}{2}) = 0.0003$) |
| $e$ ($n = 2$) | 0.12 | kinetic-dominated |

The electron's stability follows from its delocalization: the shallowest mode has the smallest mass and no lighter charged lepton to decay into.

The Koide parameter [23] $Q = (\sum\sqrt{m_n})^2/\sum m_n$ measures the balance between democratic (equal-mass) and hierarchical (mass-splitting) components of the mass spectrum. $Q = 3/2$ when these two components carry equal weight—the maximum-entropy partition for a binary question (democratic vs. hierarchical), reflecting A2. High-precision computation (§8) gives

$$Q = 1.499985 \qquad (\delta = -1.5 \times 10^{-5}). \tag{19}$$

The experimental value is $Q = 1.500014 \pm 0.000015$ (PDG 2024). The difference ($\approx 2\sigma$) is resolvable by precision tau-mass measurements at Belle II [34] (2027–2030).

## 8.1 Mass formula

Why is the $\tau$ heavy and the electron light? In the Standard Model, each charged-lepton mass is an independent parameter. Here, the three leptons are eigenstates of the same well $V = -H$: $\tau$ sits deep and is localized; $e$ sits near the rim and is spread out. In the present framework, a localized state couples more strongly to the internal gauge structure and acquires a larger mass. The mass ratios are therefore determined by the wavefunction structure—not by independent parameters. The key quantity is $R = \ln(m_\tau/m_e)/\ln(m_\mu/m_e)$, which compresses the three-body mass hierarchy into a single number. We derive it below.

$$V = -H \xrightarrow{\S 3} \psi_n \xrightarrow{\text{IPR}} \text{localisation} \xrightarrow{2C_F} m_n \to R = 1.5293 \tag{20}$$

Each eigenstate has a binding energy $|E_n|$ and a degree of localization, measured by the inverse participation ratio $\mathrm{IPR}_n = \int |\psi_n|^4 \, d\phi$. A localized state (large IPR) couples strongly to gauge fields; a delocalized state (small IPR) couples weakly. The mass scales as a power of the localization: $m_n \propto \mathrm{IPR}_n^b$. The exponent $b$ is determined by group theory: in 0D there are no loop integrals, so the anomalous dimension is purely the Casimir of the fermion bilinear under the $\mathrm{PGL}(3, \mathbb{F}_3)$ gauge group acting on $N = 3$ generation indices, $b = 2C_F = (N^2-1)/N = 8/3$.[2]

A small correction arises from the anharmonicity of $V = -H$. Each mode samples the well differently; the virial ratio $|\langle V\rangle_n|/T_n$ (potential energy to kinetic energy) measures

[2] The Casimir $(N^2-1)/N$ has the same form as the fundamental representation of SU($N$); this is not a coincidence but follows from the shared $N = 3$ structure of the generation space and the color group [20].

this. The exponent of this correction is fixed by the spectral $\zeta$-function: the WKB residue $n_{\mathrm{WKB}} = 2.781$ falls short of $N = 3$ by $\varepsilon = 0.219$, giving a deficit per mode $\delta = \varepsilon/N = 0.073$.

The complete mass formula is:

$$m_n \propto |E_n| \times \mathrm{IPR}_n^{2C_F} \times \left(\frac{|\langle V\rangle_n|}{T_n}\right)^{\varepsilon/N}. \tag{21}$$

Every exponent is determined by $V = -H$. The result:[3]

$$R = \frac{\ln(m_\tau/m_e)}{\ln(m_\mu/m_e)} = 1.5293 \qquad \text{(experiment: } 1.5294,\ 0.006\%\text{).} \tag{22}$$

The Koide constraint $Q \approx 3/2$ and the spectral condition $c = \varepsilon/N$ are simultaneously satisfied only by $b = 2C_F = 8/3$:

| $b$ | $c\,(Q{=}3/2)$ | $\lvert c - \varepsilon/N\rvert/(\varepsilon/N)$ |
|---|---|---|
| 2.0 | 0.437 | 500% |
| 2.5 | 0.163 | 120% |
| **$\mathbf{8/3 = 2C_F}$** | **0.0730** | **0.06%** |
| 2.78 | 0.011 | 85% |
| 3.0 | −0.107 | 246% |

The individual mass ratios:

| | Predicted | Experiment | Accuracy |
|---|---|---|---|
| $m_\tau/m_e$ | 3481 | 3477 | 0.1% |
| $m_\mu/m_e$ | 207.0 | 206.8 | 0.1% |
| $R$ | 1.5293 | 1.5294 | 0.006% |
| $Q$ (Koide) | 1.499985 | 1.50001(2) | $2\sigma$ (Belle II) |

The three lepton masses are not independent parameters but three modes of a single well, with exponent $2C_F = 8/3$ fixed by representation theory. In 0D there are no loop integrals, so the anomalous dimension is the Casimir eigenvalue of $\mathrm{PGL}(3, \mathbb{F}_3)$ itself. The Peter–Weyl theorem for finite groups guarantees that the heat kernel on the group takes the form $K(t) = \sum_\rho \dim(\rho)\, \chi_\rho(g)\, e^{-C_\rho t}$, admitting only exponential decay with no power-law terms. This fixes $b = 2C_F$ uniquely (rigorous proof in [20]). The 0.006% agreement of $R = 1.5293$ is a quantitative verification of this structure.

## 8.2 Quark masses

Leptons are colour singlets; quarks carry three colours. By Shannon entropy additivity, the colour-triplet potential is $V = -3H$. The deeper well supports $N = 5$ bound states ($n_{\mathrm{WKB}} = 4.82$).

**Up-type.** Mode 0 is deeply confined and belongs to the $I_3 = -1/2$ sector. Up-type quarks ($I_3 = +1/2$) use modes $(1, 2, 3)$, giving $R_{\mathrm{up}} = 1.777$ (experiment: 1.772, 0.3%).

**Down-type.** With constant mass, $R_{\mathrm{down}} = 2.096$—7.6% off experiment (2.269). The discrepancy points to a mass correction from weak isospin.

[3] All spectral quantities are computed with $|\phi|_{\max} = 60$, $\Delta\phi = 0.0002$ ($N_{\mathrm{grid}} = 512001$), verified stable under doubling of grid resolution.

The SU(2) doublet $(u, d)$ carries total color Casimir $2C_F = (N_c^2-1)/N_c$ plus baryon U(1) charge $1/N_c$. The two components are distinguished by weak isospin. Taking $I_3 = +1/2$ (up-type) as the reference, the $I_3 = -1/2$ (down-type) component receives an effective mass correction from the combined color and baryon charges. Weak-isospin reversal couples all gauge charges to the internal Fisher metric $\sigma(1-\sigma)$. The mechanism originates from the geometry of $\mathrm{PG}(2, \mathbb{F}_3)$: among the $N+1 = 4$ points on line $L$, the reference point $P_0$ defines $I_3 = +1/2$, and the remaining 3 points generate SU(2). The isospin flip $P_0 \leftrightarrow L\backslash P_0$ reverses the SU(2) action, coupling the total charge $N_c$ to $\sigma(1-\sigma)$ with a negative sign (uniqueness proof in Paper II, Theorem 5). The sum $2C_F+1/N_c = N_c$ is an exact algebraic identity, giving an effective mass:

$$m_{\text{eff}}(\phi) = 1 - N_c\,\sigma(1-\sigma) \tag{23}$$

for $I_3 = -1/2$, while $m_{\text{eff}} = 1$ for $I_3 = +1/2$ (up-type unaffected). With $g = -N_c = -3$ and modes $(0, 1, 2)$:

$$R_{\text{down}} = 2.273 \qquad \text{(experiment: } 2.269,\ 0.2\%\text{)}. \tag{24}$$

The 7.6% discrepancy improves to 0.2%. The weak-isospin correction is not an ad hoc adjustment but is uniquely determined by the algebraic structure of the SU(2) doublet.

All three charged-fermion mass ratios are reproduced:

| Sector | $c$ | $g$ | $R$ (predicted) | $R$ (experiment) |
|---|---|---|---|---|
| Lepton | 1 | 0 | 1.529 | 1.529 (0.006%) |
| Up | $N_c = 3$ | 0 | 1.777 | 1.772 (0.3%) |
| Down | $N_c = 3$ | $-N_c$ | 2.273 | 2.269 (0.2%) |

The mode assignments—up $= (1, 2, 3)$, down $= (0, 1, 2)$—follow from the deeper well at $c = N_c = 3$: mode 0 is deeply confined, and the $I_3 = -1/2$ component with $g = -N_c$ supports $N = 4$ states, of which $(0, 1, 2)$ are the three lightest.

## 8.3 CKM mixing

Why is $V_{ub}$ small, and why does the Cabibbo angle [19, 24] have the value it does? The Standard Model offers no explanation. In $V = -H$, the potential symmetry determines the former and the marginality of the third generation determines the latter.

The potential $V = -H$ is symmetric: $V(\phi) = V(-\phi)$. This forces the eigenfunctions into definite parity: $\psi_0$ even, $\psi_1$ odd, $\psi_2$ even. The transition matrix element between modes 0 and 2 vanishes by parity:

$$V_{ub}\big|_{\text{tree}} = 0 \qquad \text{(numerically: } 1.9 \times 10^{-13}\text{)}. \tag{25}$$

The physical value $|V_{ub}| = 0.004$ arises at higher order through the Wolfenstein hierarchy $|V_{ub}| \sim \varepsilon^3$ (see below).

The Cabibbo angle follows from the WKB deficit. $\varepsilon = 0.219$ measures how marginal the electron is: $\varepsilon \to 0$ eliminates it, reducing to two generations and zero mixing. To second order:

$$\sin\theta_C = \varepsilon\Big(1 + \frac{\varepsilon}{N^2}\Big) = 0.2245 \qquad \text{(experiment: } 0.2244,\ 0.05\%\text{)}. \tag{26}$$

The Cabibbo angle is a direct expression of the fact that the third generation barely exists: $\varepsilon = 0.219$. Were the electron bound slightly more deeply ($\varepsilon \to 0$), inter-generation mixing would vanish and all flavour transitions would be forbidden.

From the Cabibbo angle, the full Wolfenstein hierarchy follows [25]: $|V_{us}| \sim \varepsilon$, $|V_{cb}| \sim \varepsilon^2$, $|V_{ub}| \sim \varepsilon^3$ (experiment: 0.225, 0.041, 0.004). The NLO corrections take the form $(1 + \varepsilon/k)$, where $k$ is determined by the $\mathrm{PG}(2, \mathbb{F}_3)$ geometry: $k = N^2 = 9$ for generation mixing ($\sin\theta_C$), $k = N^2{+}1 = 10$ for the non-weak sector ($\alpha_s$), and $k = N = 3$ for the reactor angle ($\sin^2\theta_{13}$). The systematic derivation of $k$-values (Borel filtration) is given in Paper II. The systematic derivation of these $k$-values from the Borel filtration of $\mathrm{PGL}(3, \mathbb{F}_3)$ is proved in [20].

The parity symmetry $V(\phi) = V(-\phi)$ that gives $V_{ub} \approx 0$ also solves a different problem: the strong CP puzzle. $\phi \to -\phi$ is particle–antiparticle interchange in occupation space; the Lagrangian inherits this symmetry:

$$\theta_{\mathrm{QCD}} = 0 \qquad \text{(experiment: } |\theta| < 10^{-10}). \tag{27}$$

The strong CP problem is resolved by $V$-parity, requiring neither the Peccei–Quinn mechanism nor an axion. Yet CP violation *is* observed in the CKM sector ($J = 3.06{\times}10^{-5}$). This does not contradict $\theta_{\mathrm{QCD}} = 0$: the CKM phase arises at higher order in $\varepsilon$ ($J = \pi\varepsilon^5/N_{\mathrm{flags}}$, Paper II §8.1), while the strong CP angle remains exactly zero.

## 8.4 Neutrinos

**Mass predictions.** Neutrino masses are the least constrained parameters in the Standard Model: the absolute scale remains unknown. $V = -H$ eliminates this freedom—the same well, the same exponent, the same $R = 1.529$ as charged leptons determine $m_1$ and the mass ordering uniquely.

Since neutrinos are color-singlets, they couple to $V = -H$ with $c = 1$—the same potential as charged leptons. The mass ratio $R_\nu = R_{\mathrm{lepton}} = 1.529$ follows. Combined with oscillation data [28] ($\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV$^2$, $\Delta m^2_{31} = 2.534 \times 10^{-3}$ eV$^2$, normal ordering), $R_\nu = 1.529$ determines $m_1$ uniquely:

| | Prediction | Status |
|---|---|---|
| $m_1$ | $0.31 \pm 0.03$ meV[4] | unmeasured |
| $m_2$ | 8.66 meV | consistent |
| $m_3$ | 50.3 meV | consistent |
| $\sum m_i$ | $59.3 \pm 0.3$ meV | below Planck ($< 120$) |
| Ordering | normal | JUNO (2026–27) |

The sum $\sum m_i = 59.3$ meV is dominated by $m_3 \approx \sqrt{\Delta m^2_{31}} = 50.3$ meV; the theory's contribution is $m_1 \approx 0.3$ meV, placing $\sum m_i$ near its minimum allowed value. The Koide parameter $Q_\nu = 1.90 \neq 3/2$: neutrinos do not satisfy the equipartition condition, consistent with the theory's prediction that $Q \approx N/2$ holds approximately only for charged leptons (all $N = 3$ modes used).

This prediction is falsifiable: inverted ordering falsifies the theory, $\sum m_i > 70$ meV creates tension, and CMB-S4 [27] ($\sigma \sim 15$ meV) can test $\sum m_i = 59.3$ meV directly. The derivation of $\Delta m^2_{31}/\Delta m^2_{21} = |\mathrm{PG}|^2/(N{+}2) = 33.8$ and $m_3/m_2 = |\mathrm{PG}|/\sqrt{N{+}2} = 13/\sqrt{5}$ from the flag Laplacian spectrum is given in Paper II.

**Mixing angles.** Unlike CKM mixing, which is small ($\sin\theta_C \approx 0.22$), PMNS mixing angles are large ($\sin^2\theta_{12} \approx 1/3$). Their origin is one of the outstanding puzzles of flavor physics. Here we derive all three PMNS angles $\theta_{12}$, $\theta_{23}$, $\theta_{13}$ from the geometry of $\mathrm{PG}(2, \mathbb{F}_3)$.

CKM mixing was controlled by the internal coordinate $\phi$ ($\varepsilon \ll 1$, perturbative). PMNS mixing is controlled by the gauge geometry ($1/N \sim 1$, non-perturbative).

A line in $\mathrm{PG}(2, 3)$ contains $N+1 = 4$ points. The solar mixing angle measures the fraction of PG covered by one complete electroweak line ($\mathrm{SU}(2) \times \mathrm{U}(1)$, including hypercharge):

$$\sin^2\theta_{12} = \frac{N+1}{|\mathrm{PG}|} = \frac{4}{13} = 0.3077 \qquad \text{(experiment: } 0.307,\ 0.2\%). \tag{28}$$

Compare $\sin^2\theta_W = N/|\mathrm{PG}| = 3/13$ (pure SU(2), excluding hypercharge); the difference $1/|\mathrm{PG}|$ is the $\mathrm{U}(1)_Y$ contribution.

The atmospheric mixing is maximal ($\mu$–$\tau$ symmetry) plus a correction at the PG scale:

$$\sin^2\theta_{23} = \frac{1}{2} + \frac{1}{2|\mathrm{PG}|} = \frac{7}{13} = 0.538 \qquad \text{(LO)}. \tag{29}$$

The NLO Borel correction with $k = |\mathrm{PG}| = 13$:

$$\sin^2\theta_{23} = \frac{7}{13}\Big(1 + \frac{\varepsilon}{13}\Big) = 0.5475 \qquad \text{(second octant)}. \tag{30}$$

The correction $1/(2|\mathrm{PG}|)$ is the minimum $\mu$–$\tau$ symmetry breaking from the gauge geometry. The sign ($> 1/2$) follows from $\tau$ being heavier (more localized, larger PG overlap), predicting the second octant ($\sin^2\theta_{23} > 0.5$). Current data do not resolve the octant: NuFIT 6.0 [28] reports $\sin^2\theta_{23} = 0.47$–$0.56$ depending on the atmospheric data included, with $3\sigma$ range 0.43–0.60. The prediction 0.5475 lies within $1.1\sigma$ of the upper-octant solution. The octant determination by DUNE and Hyper-Kamiokande ($\sim$2030) will provide a decisive test.

Consistency check: $\sin^2\theta_{23} = \sin^2\theta_W + \sin^2\theta_{12}$ ($7/13 = 3/13 + 4/13$). Atmospheric mixing exhausts both the pure-weak ($N$ points) and the full electroweak-line ($N+1$ points) fractions of $\mathrm{PG}(2, 3)$.

The reactor angle is suppressed by the full PG: $\sin^2\theta_{13} = 1/(N \cdot |\mathrm{PG}|) = 1/39 = 0.026$ at leading order. The NLO Borel correction at the center $Z(U)$ ($k = N = 3$) gives $\sin^2\theta_{13} = (1/39)(1 - 2\varepsilon/3) = 0.0219$ (experiment: 0.02195, 0.3%). NLO corrections to all three PMNS angles and the CKM CP phase are derived in [20].

**Dirac particles.** Whether neutrinos are Dirac or Majorana is an open experimental question. The framework makes a definite prediction. Two independent arguments establish that neutrinos are Dirac, not Majorana. First, each generation is a single mode with $n \in \{0, 1\}$. A Majorana mass term identifies a particle with its own antiparticle ($\nu = \nu^c$), but in the single-mode framework each bound state carries a definite generation quantum number that distinguishes $\nu$ from $\nu^c$; the self-conjugate identification has no room in the $n \in \{0, 1\}$ structure. Second, every fermionic operator entering the construction is a function of the number operator $\hat{n}$ — the sufficient statistic of the Bernoulli family — and is therefore invariant under the phase rotation $a \to e^{i\theta}a$: the anomaly-free combination $\mathrm{U}(1)_{B-L}$ is exact, and the Majorana bilinear $aa + a^\dagger a^\dagger$ ($\Delta(B-L) = 2$) lies outside the operator algebra of the framework. V-parity then pairs the distinct states $\nu$ and $\nu^c$ ($L = \pm 1$) into a Dirac fermion (operator-level proof: Paper II, Theorem 9 [20]). Neutrinoless double beta decay is therefore absent ($0\nu\beta\beta = 0$), testable by nEXO [29] ($\sim$2030).

This completes the derivation of the Standard Model parameters—mass ratios, mixing angles, gauge couplings, the Higgs mass ratio, and neutrino predictions—all from $V = -H$ and $\mathrm{PG}(2, \mathbb{F}_3)$. The Standard Model treats these as independent free parameters; in the present framework, they are all different aspects of a single equation.

# 9 Connection to Spacetime

All calculations in §2–7 were performed in internal space (the $\phi$ coordinate). Neither space nor time appeared. How, then, does this zero-dimensional theory connect to four-dimensional spacetime? This chapter answers two questions: why spacetime is four-dimensional with signature $(3, 1)$ (§9.1), and why internal space and spacetime are independent (§9.2).

## 9.1 Spacetime dimension and gravity

Section 4 derived $d_{\text{space}} = 3$. But "why four-dimensional spacetime?" is a separate question—it is not obvious that time is one-dimensional. We show that two independent paths give the same $d = 4$.

The first path comes from gravitational degrees of freedom. A massless spin-2 particle (graviton) in $d$-dimensional spacetime has $d(d-3)/2$ physical degrees of freedom (from Poincaré-group representation theory, without assuming any particular $d$ or Einstein's equations). The $N = 3$ bound states of $V = -H$ provide $N-1 = 2$ independent internal degrees of freedom (three probabilities sum to one). Under the tensor-product structure (§9.2), spacetime dynamical degrees of freedom must match internal ones:

$$d(d-3)/2 = N - 1 = 2 \quad \Longrightarrow \quad d = 4. \tag{31}$$

The second path comes from projective geometry. Section 5.1 lifted $\mathrm{PG}(2, \mathbb{F}_3)$ to a continuous gauge group. Applying the same lift to the full geometry, the algebraic closure $\mathbb{F}_3 \to \mathbb{C}$ gives $\mathrm{PG}(2, \mathbb{C}) = \mathbb{C}P^2$. $\mathbb{C}P^2$ is a real 4-dimensional manifold:

$$d = \dim_{\mathbb{R}}(\mathbb{C}P^{N-1}) = 2(N-1) = 4. \tag{32}$$

The two paths are summarised:

| Path | Input | Equation | Result |
|---|---|---|---|
| Graviton DOF | Spin-2 polarisations | $d(d-3)/2 = N-1 = 2$ | $d = 4$ |
| Projective geometry lift | $\mathbb{F}_3 \to \mathbb{C}$ | $\dim_{\mathbb{R}}(\mathbb{C}P^{N-1}) = 2(N-1)$ | $d = 4$ |

These two paths are independent—one from field-theoretic representation theory (spin-2 polarisations), the other from pure algebraic geometry ($\mathbb{F}_3 \to \mathbb{C}$). That both give $d = 4$ is a non-trivial consistency check of $N = 3$.

Even with $d = 4$ fixed, the signature could be $(4, 0)$, $(3, 1)$, or $(2, 2)$. $\mathbb{F}_3 = \{0, 1, 2\}$ does not contain $\sqrt{-1}$ ($1^2 = 1$, $2^2 = 1$, so $x^2 = -1 \equiv 2$ has no solution). The minimal extension supplying it is $\mathbb{F}_9 = \mathbb{F}_3(i)$. This $i$ distinguishes one direction of $\mathbb{C}^2 \cong \mathbb{R}^4$ as "imaginary." Furthermore, the one-dimensional internal coordinate $\phi$ opened by A2 (§2.2) connects to the time dimension of spacetime as the evolution parameter $\tau$ in the action $S = \int [\frac{1}{2}\dot{\phi}^2 + V]\, d\tau$. The full logic gives signature $(3, 1)$:

$$\mathbb{F}_3 \xrightarrow{\text{no } \sqrt{-1}} \mathbb{F}_9 = \mathbb{F}_3(i) \xrightarrow{\text{lift}} \mathbb{C}^2 \cong \mathbb{R}^4 \xrightarrow{i \text{ marks 1 axis}} (3, 1). \tag{33}$$

In four dimensions, Lovelock's theorem [30] uniquely determines the gravitational field equation: $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$. Jacobson [31] showed that the same equation follows from $\delta Q = T_H\, dS$ on a causal horizon. Since the entropy of the present theory is $S = H(P)$, both paths lead from binary entropy to Einstein gravity.

## 9.2 Why $\phi$ and $x$ are independent

The coordinate $\phi = \mathrm{logit}(\sigma)$ lives in internal space; the coordinate $x$ lives in spacetime. Why are they independent?

The answer is $n^2 = n$. This algebraic identity holds independently of spatial position—$0^2 = 0$ and $1^2 = 1$ everywhere in the universe. Since $V = -H(\sigma(\phi))$ is constructed from $n^2 = n$ alone, $\partial V/\partial x = 0$ holds exactly. No coupling between $\phi$ and $x$ exists, and the full Hamiltonian is $H = H_\phi + H_x$. The Hilbert space decomposes as $\mathcal{H}_{\mathrm{gen}}(\phi) \otimes \mathcal{H}_{\mathrm{space}}(x)$. The tensor product is not an assumption but a consequence of $n^2 = n$:

$$n^2 = n \;\Rightarrow\; V = V(\phi) \;\Rightarrow\; \frac{\partial V}{\partial x} = 0 \;\Rightarrow\; H = H_\phi + H_x \;\Rightarrow\; \mathcal{H}_{\mathrm{gen}}(\phi) \otimes \mathcal{H}_{\mathrm{space}}(x). \quad (34)$$

Since this composite space has dim $\geq 3$ ($\mathcal{H}_{\mathrm{gen}}(\phi)$ is 3-dimensional, $\mathcal{H}_{\mathrm{space}}(x)$ is infinite-dimensional), Gleason's theorem [32] uniquely extends the single-mode detection probability $\sigma = \langle n \rangle$ of §2 to the standard Born rule—the measurement rule is not an axiom but a consequence of the tensor-product structure.

Two observational facts confirm this. First, all gauge couplings are independent of generation—direct evidence that $\phi$ and $x$ are not coupled. Second, the 0D theory computes only dimensionless quantities (mass ratios, mixing angles), while dimensionful quantities (absolute masses) require spacetime—precisely the prediction of the tensor-product structure.

The analogy with spin illuminates this structure. Spin algebra is finite-dimensional and independent of the Lagrangian, yet determines 4D physical quantities (magnetic moment, fine structure). Generations have the same structure—the finite-dimensional $V = -H$ determines 4D mass ratios and mixing angles.

## 9.3 Correspondence between 0D values and 4D experiment

The tensor-product structure explains why 0D algebraic values coincide with tree-level values of the 4D renormalised theory. Since 0D has no energy scale, the 0D values are scale-invariant algebraic constants. In 4D, tree level is the limit of zero loop corrections—i.e. the limit in which spacetime dynamics does not affect internal structure—which is precisely the independence guaranteed by $\mathcal{H}_{\mathrm{int}}(\phi) \otimes \mathcal{H}_{\mathrm{space}}(x)$. Radiative corrections arise from loop momenta in the spacetime factor and enter as sub-percent corrections to the internal algebraic values (§7: $\sin^2\theta_W = 3/13 + \Delta_{\mathrm{rad}} = 0.23122$). The 0D values are neither RG fixed points nor UV boundary conditions, but the unique algebraically determined point in 4D parameter space—a value protected from 4D loop corrections by the tensor-product structure.

With this, the relationship between internal structure ($\phi$) and spacetime ($x$) is established. $V = -H$ determines mass ratios, mixing angles, and coupling constants, while spacetime supplies only the overall energy scale.

# 10 Results

From the single equation $V = -H$ and the single geometric structure $\mathrm{PG}(2, \mathbb{F}_3)$, we have shown that 20 of the 26 free parameters of the Standard Model are determined to 0.0006%–2.4% accuracy with zero adjustable parameters. The remaining 6 ($|V_{cb}|$, $|V_{ub}|$, $\delta_{\mathrm{CKM}}$, $\delta_{\mathrm{PMNS}}$, $m_3/m_2$, $\Delta m^2_{31}/\Delta m^2_{21}$) require the Borel filtration structure and restricted flag Laplacian spectra, and are derived in Paper II.

The Standard Model treats these quantities as independent inputs, yet the present derivation contains no free parameters. The agreement across multiple independent quantities at this precision strongly suggests that this construction is valid. The full list of results is given below.

**Table 1.** Comparison with Standard Model constants (experimental values from PDG 2024 [22], NuFIT 6.0 [28], and CODATA 2022 [35]).

| Parameter | Prediction | Experiment | Accuracy | § |
|---|---|---|---|---|
| $1/\alpha_{\mathrm{em}}$ | 137.0368 | 137.036 | 0.0006% | 6.3 |
| $\alpha_s$ | 0.1179 | 0.1180 | 0.08% | 6.2 |
| $\sin^2\theta_W$ ($\to g_1/g_2$) | 3/13 | 0.23122 | 0.2% | 6.1 |
| $m_H/v$ ($\to \lambda$) | 0.5084 | 0.5087 | 0.07% | 7 |
| $R_{\mathrm{lepton}}$ ($\to y_e, y_\mu, y_\tau$) | 1.5293 | 1.5294 | 0.006% | 8.1 |
| $R_{\mathrm{up}}$ ($\to y_u, y_c, y_t$) | 1.777 | 1.772 | 0.3% | 8.2 |
| $R_{\mathrm{down}}$ ($\to y_d, y_s, y_b$) | 2.273 | 2.269 | 0.2% | 8.2 |
| $\sin\theta_C$ | 0.2245 | 0.2244 | 0.05% | 8.3 |
| $\sin^2\theta_{12}$ | 4/13 | 0.307 | 0.2% | 8.4 |
| $\sin^2\theta_{23}$ | 0.5475 | 0.561 | 2.4% | 8.4 |
| $\sin^2\theta_{13}$ | 0.0219 | 0.02195 | 0.3% | 8.4 |
| $m_1$ | 0.31 meV | — | prediction | 8.4 |
| $\theta_{\mathrm{QCD}}$ | 0 | $< 10^{-10}$ | exact | 8.3 |

**Table 2.** Other derived results.

| Quantity | Derivation | Experiment | § |
|---|---|---|---|
| $N$ (generations) | $n_{\mathrm{WKB}} = 2.781 \to 3$ | 3 | 3 |
| $d$ (spatial) | $\{1,3\} \cap \{\geq 3\} \cap \{\geq 3\}$ | 3 | 4 |
| $d$ (spacetime) | $d(d-3)/2 = N-1 = 2$ | 4 | 9.1 |
| Signature | $\mathbb{F}_3(i) \to \mathbb{C}P^2$ | $(3, 1)$ | 9.1 |
| Gauge group | 9+3+1 (Killing–Cartan) | SU(3)×SU(2)×U(1) | 5 |
| Spectral gauge | Restricted Laplacian products 9+4=13 | — | Paper II |
| 4th generation | $E_3 > 0$ | excluded | 3.2 |
| Neutrino nature | $\mathrm{U}(1)_{B-L}$ ($\hat{n}$ structure) + $V$-parity | — | 8.4 |
| $m_W/m_Z$ | $\sqrt{10/13} \to +\Delta\rho$: 0.8812 | 0.8814 (0.02%) | 6.1; 7 |
| $Q$ (Koide) | 1.499985 | 1.50001 | 8.1 |
| $\sin^2\theta_W$ + 1-loop | 0.23122 | 0.23122 ($<$ 0.01%) | 7 |

**Table 3.** Testable predictions.

| Prediction | Value | Test | § |
|---|---|---|---|
| Mass ordering | Normal | JUNO [26] (2026–27) | 8.4 |
| $\Delta m^2_{31}/\Delta m^2_{21}$ | $\lvert PG\rvert^2/(N{+}2) = 33.8$ | JUNO [26] (2026–27) | Paper II |
| $\theta_{23}$ octant | Second ($> 45°$) | DUNE / HK (∼2030) | 8.4 |
| $\delta_{\mathrm{PMNS}}$ | 197.1° | DUNE / HK (∼2030) | Paper II |
| $\Sigma m_\nu$ | 59.3 meV | CMB-S4 [27] (∼2030) | 8.4 |
| $0\nu\beta\beta$ | Strictly zero | nEXO [29] (∼2030) | 8.4 |
| Koide deviation $\delta$ | $-1.5 \times 10^{-5}$ | Belle II [34] (2027–30) | 8.1 |
| No BSM particles | — | LHC / FCC | 3 |

# 11 Conclusion

This paper has shown that the results of mathematical development and physical insight derived from the axioms of existence agree with experimental values to high precision, demonstrating that all matter is different aspects of a single equation

$$V = -H(\sigma(\phi)), \tag{35}$$

and that they are not independent. This suggests that every fundamental law of nature can be described, without adjustment and as a necessity, from this equation derived from the axioms of existence. Paper II [20] provides the derivation of the remaining 6 parameters—including the neutrino mass ratio $m_3/m_2 = |\mathrm{PG}|/\sqrt{N{+}2}$ from the flag Laplacian spectrum, and the PMNS CP phase $\delta_{\mathrm{PMNS}} = 2\pi\sin^2\theta_{23}$ as a Berry phase—together with the spectral proof of the gauge decomposition 9+3+1 via restricted Laplacians, and more rigorous uniqueness proofs of the constructions employed in this paper. Cosmological consequences are developed in Paper III [33].

*To be, or not to be — that is the equation.*

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